

Proposal in Response to RFQ – Solar-Powered Irrigation Systems

Submitted to: Supremo Green Revolution (SGR)

Submitted by: SunHarvest Solutions Ltd. (SHS)

Date: October 7, 2025

1. Executive Overview

SunHarvest Solutions Ltd. (SHS) is pleased to present this proposal in response to the RFQ issued by Supremo Green Revolution (SGR) for the provision of solar-powered irrigation systems in Kenya, Ghana, and Tanzania.

Founded in 2020, SHS is a young enterprise driven by the mission to make renewable energy accessible to underserved farming communities. Our focus is on simple, low-cost, and easy-to-operate solar irrigation systems suitable for smallholder farmers.

While SHS is still scaling its operations, we believe that our local partnerships, community engagement efforts, and innovative financing ideas make us a valuable partner for SGR's Green Resilience Initiative.

2. Organizational Profile

SunHarvest Solutions Ltd. is registered in Nairobi, Kenya, with partner field representatives in Tanzania and Ghana. The company currently employs 18 full-time staff, including 3 engineers, 2 agronomists, and 5 field coordinators. Our management team has prior experience in NGO and donor-funded projects but limited large-scale technical delivery.

Although SHS is not yet ISO-certified, the company follows basic quality assurance procedures and is currently in the process of applying for ISO 9001 certification in 2026. Environmental and occupational safety standards are managed through internal policies.

Key partnerships include local cooperatives, youth groups, and rural microfinance institutions supporting farmer-led irrigation adoption.

3. Technical Proposal

SHS proposes to deliver small, modular solar irrigation systems designed primarily for farms between 0.25 and 2 hectares. The systems are intended for basic crop irrigation and livestock watering in areas with shallow groundwater access.

****System Specifications:****

- Solar PV array (1.5–3.0 kW, polycrystalline modules).
- DC surface pump (capacity: 3,000–6,000 L/day).
- Steel support frame with manual tilt adjustment.
- 2,000–3,000 L plastic water storage tanks.
- Simple drip kits for horticultural crops (optional).

****Implementation Approach:****

1. Site identification in collaboration with local cooperatives.

- 2. Procurement from regional suppliers (primarily in Nairobi).
- 3. Local assembly and installation with support from trained youth technicians.
- 4. Farmer orientation and demonstration sessions at installation sites.

SHS does not currently use digital monitoring systems but intends to introduce low-cost IoT pump sensors during the pilot phase.

4. Deliverables

- 1. Nine (9) complete solar irrigation installations (3 per country).
- 2. Basic user training sessions for approximately 450 farmers.
- 3. Printed manuals in English and Swahili (translations to other languages pending).
- 4. One-year limited warranty on pumps and panels.
- 5. Basic quarterly reports summarizing operational performance and farmer feedback.

5. Quality Assurance and Compliance

SunHarvest Solutions follows internally developed Standard Operating Procedures (SOPs) for equipment inspection, installation safety, and customer training. However, SHS does not currently hold ISO or IEC certification. Vendor-supplied components are sourced from ISO-compliant manufacturers.

Environmental and occupational safety are monitored informally, and the company acknowledges the need to develop formal compliance structures as operations expand.

6. Implementation Timeline

Phase	Activity	Duration
1	Site Survey & Selection	2 weeks
2	Procurement & Shipping	4 weeks
3	Installation	4 weeks
4	Training & Handover	2 weeks
5	Monitoring & Reporting	Ongoing

Total Implementation Duration: **12 weeks** (subject to logistics and customs delays).

7. Financial Proposal (Indicative, USD)

Item	Quantity	Unit Cost (USD)	Subtotal (USD)
Solar Irrigation System (basic package)	9	4,500	40,500
Farmer Training	9	1,500	13,500
Spare Parts & Maintenance Kits	9	300	2,700
Logistics & Local Transport	1	8,000	8,000
Administration & Reporting	1	6,500	6,500
Total (Excl. VAT)			**71,200**

Notes:

- Costs are preliminary estimates and subject to revision.
- SHS cannot guarantee uniform pricing due to volatile import duties.

- Warranty coverage limited to 12 months post-installation.

8. Sustainability and After-Sales Service

SHS plans to establish regional service partnerships for maintenance and spare parts distribution. A small portion of revenues will be reinvested into farmer cooperatives for system upkeep.

Key initiatives include:

- “Train-the-trainer” model to build local technician capacity.
- Exploration of mobile payment systems for service requests.
- Ongoing engagement with youth groups to promote renewable irrigation adoption.

However, SHS currently lacks a formalized monitoring platform or centralized support line, which may limit response speed for technical issues.

9. Strengths and Weaknesses

****Strengths:****

- Cost-effective and simple systems ideal for smallholder entry-level irrigation.
- Strong grassroots relationships with cooperatives and youth organizations.
- Willingness to co-develop systems under SGR technical supervision.

****Weaknesses:****

- Lack of ISO and IEC certifications.
- Limited technical workforce and absence of monitoring infrastructure.
- Weak financial reserves and dependency on external funding.
- One-year warranty below SGR’s minimum requirement (24 months).
- Minimal documentation and data management capacity.

10. Conclusion

SunHarvest Solutions appreciates the opportunity to collaborate with Supremo Green Revolution (SGR) under the Green Resilience Initiative. While the company acknowledges several capacity gaps — particularly in certification, quality assurance, and monitoring — SHS remains deeply committed to learning, partnership, and innovation at the community level.

We believe our grassroots experience, low-cost product design, and willingness to adapt provide a meaningful foundation for pilot collaboration and future scaling with SGR’s guidance and technical support.

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